

Data Cleaning Guide

"Garbage in, garbage out."

Poor labels cause **false positives** during prediction, while clean labels help the model learn correctly.

A **false positive** means the model says, "**I found something,**" where there is actually nothing useful.

If we train the model using poor labels, the model will learn poor behavior. A bounding box is not just a rectangle; it is an instruction to the AI.

Each .txt label file contains the class and coordinates of the labeled objects in the corresponding image. Therefore, if the coordinates are wrong, too large, or misplaced, the model will learn the wrong location and object shape.

Please make sure every box is **tight, accurate, and meaningful** before training.

Before training, check the following:

1. Meter bbox

The meter box should cover the full physical water meter only, not the surrounding background.

2. Window bbox

The window box must tightly cover only the digit-display area.

Do not include screws, circular dials, empty plastic space, the meter body, or background.

Poor window labels directly cause false positives during prediction. For example, if the window box includes screws, circular dials, empty plastic, or meter body parts, the model may later detect those areas as a reading window.

3. Digit bboxes

Each digit box must be tight around the actual digit only. Avoid large boxes that include unnecessary space.

4. Unknown class

Use unknown only when a digit is unclear, hidden, blurred, or impossible to read.

5. Poor images

Remove an image from the training dataset if it is too blurry, too dark, badly cropped, or impossible to label correctly.

Some cases:



meter

window

0	0	1	7	8	8	9	6
0	0	1	7	8	8	9	6

- bbox bottom looks inaccurate
- window box is too large
- some bboxes contain unnecessary spaces
e.g. the first digit left (0)

meter

window

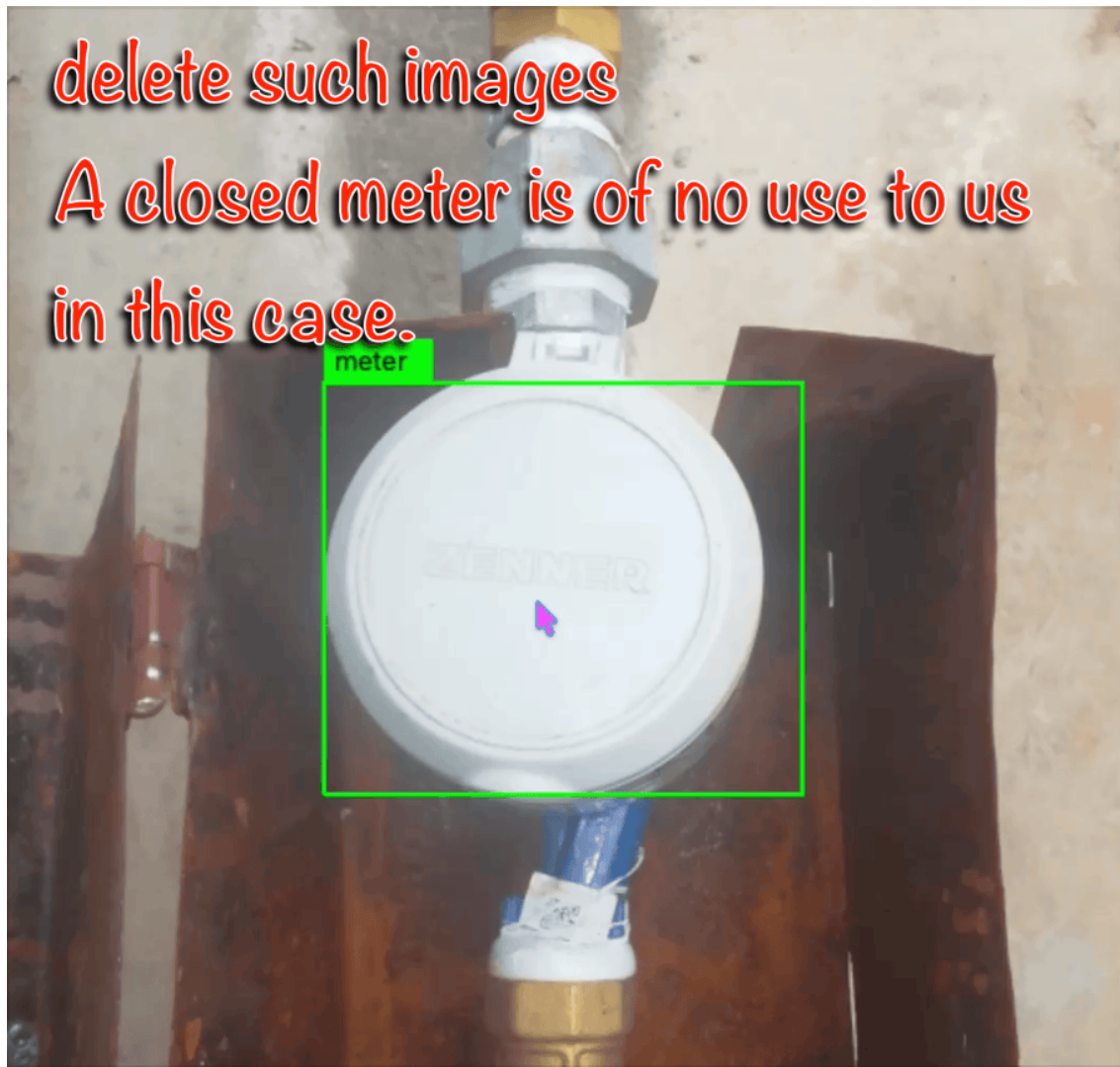
0	0	1	7	8	8	9	6
0	0	1	7	8	8	9	6

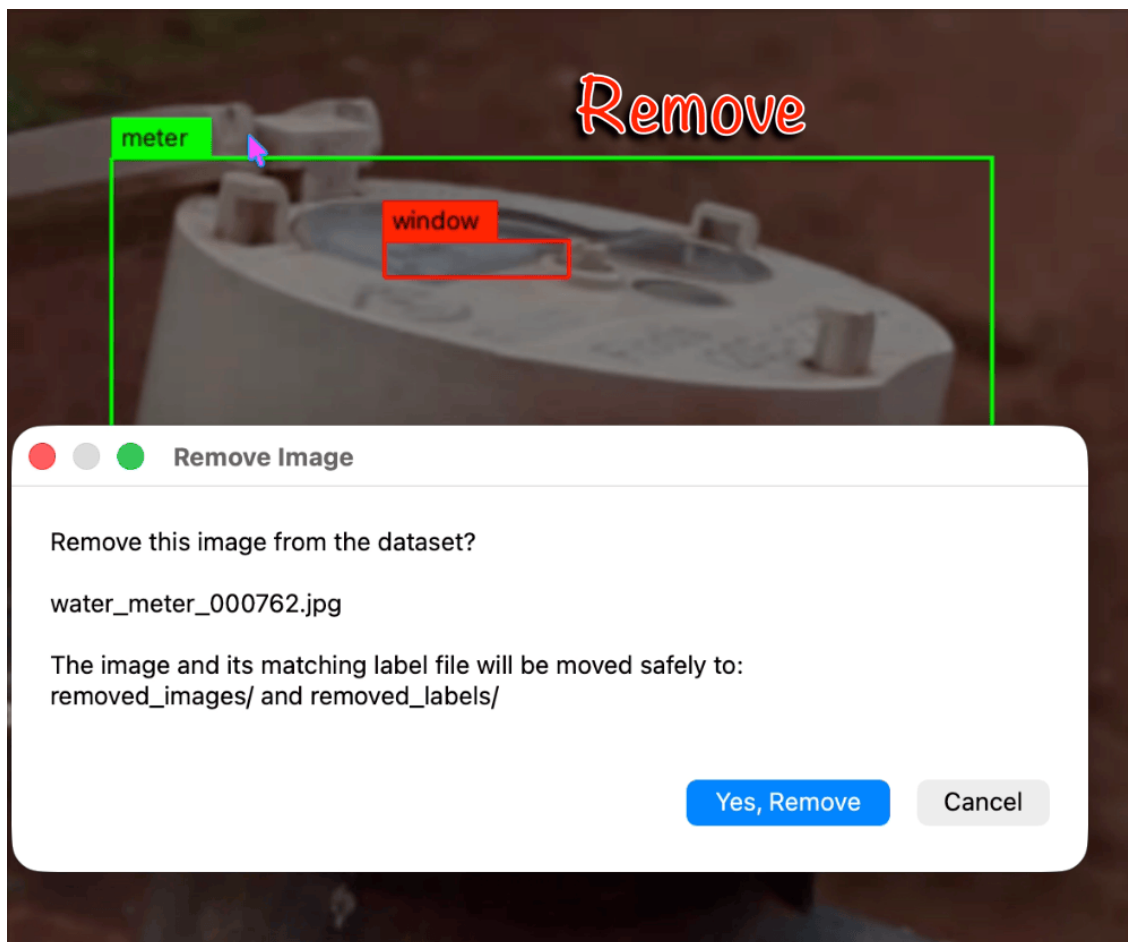
- bbox bottom looks inaccurate
- meter box is too large
- some bboxes contain unnecessary spaces
e.g. the first digit left (0)



delete such images

A closed meter is of no use to us
in this case.





meter

Remove

window

Remove Image

Remove this image from the dataset?

water_meter_000762.jpg

The image and its matching label file will be moved safely to:
removed_images/ and removed_labels/

Yes, Remove

Cancel